How dissipation engineering shapes the memory of quantum reservoirs

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Quantum reservoir computers process time series through the fading memory of a dissipative quantum system. We show that the shape of the dissipation, set by its jump operators, matters as much as its rate. At equal total coupling, spins that decay independently forget faster as the loss grows, whereas spins sharing one decay channel forget more slowly: strong loss holds them in dark states through the quantum Zeno effect. A long memory has a price, because later inputs scramble earlier ones into products that a linear readout cannot use. For a class of reservoirs we prove that the number of inputs mixed into a stored memory is exponentially distributed, with mean equal to how hard one input nudges it times how long it lasts. The input must therefore weaken as the memory grows, which more than doubles the linear memory under exact readout. With finite sampling, weak local loss, which cools the spins, retains the most memory; dark-state memories last as long but are over ten times fainter.

1 Introduction

Throw pebbles into a pond, and the ripples hold a fading record of the last few throws. Reservoir computing turns this into a machine [1, 2]: a fixed dynamical system, the reservoir, is driven by a stream of inputs, and only a linear readout of its state is trained, to recall past inputs or to predict what comes next; even a bucket of water has served as a reservoir [3]. A good reservoir holds on to recent inputs and lets go of old ones and of its own starting state. In quantum reservoir computing (QRC) the system is quantum [4], here a few interacting spins whose expectation values are read out. Unitary dynamics never forgets, so the letting go must come from the environment [5]: damping, absorption, measurement and resets all create or regulate fading memory [6–15].

The usual knob is the loss rate. Tuning how fast each qubit relaxes improves a driven spin reservoir [7], memory peaks at an intermediate decay that coincides with maximal optical absorption [9], and drive, interactions and damping have been tuned together [16], also by us [17]. In this picture dissipation is a dial with a sweet spot: the more the reservoir leaks, the faster it forgets.

But dissipation can be designed, not just dialled. The jump operators decide how a reservoir leaks: each spin into its own channel, or all spins through one shared channel, in which the decay paths interfere and some collective states, the dark states, cannot decay at all, as in subradiance [18–20]. Choosing jump operators is the craft of dissipation engineering, which prepares and protects quantum states and even computes with them [21–24]. Earlier QRC studies tuned the rates of local channels [7], compared local noise types in gate-based reservoirs [25, 26], found that stronger photon loss favours redundant encoding in coupled Kerr oscillators [27], or read a static classifier out of a decoherence-free subspace [28]. Here we ask how the shape of the dissipation, not only its strength, sets the memory of a quantum reservoir.

We find three things, which become three design rules (Fig. 1). First, the jump operators set the direction of the memory clock: at the same total coupling, spins that leak on their own forget faster as the loss grows, while spins that share one channel forget more slowly, because strong loss holds the reservoir in its dark states through the

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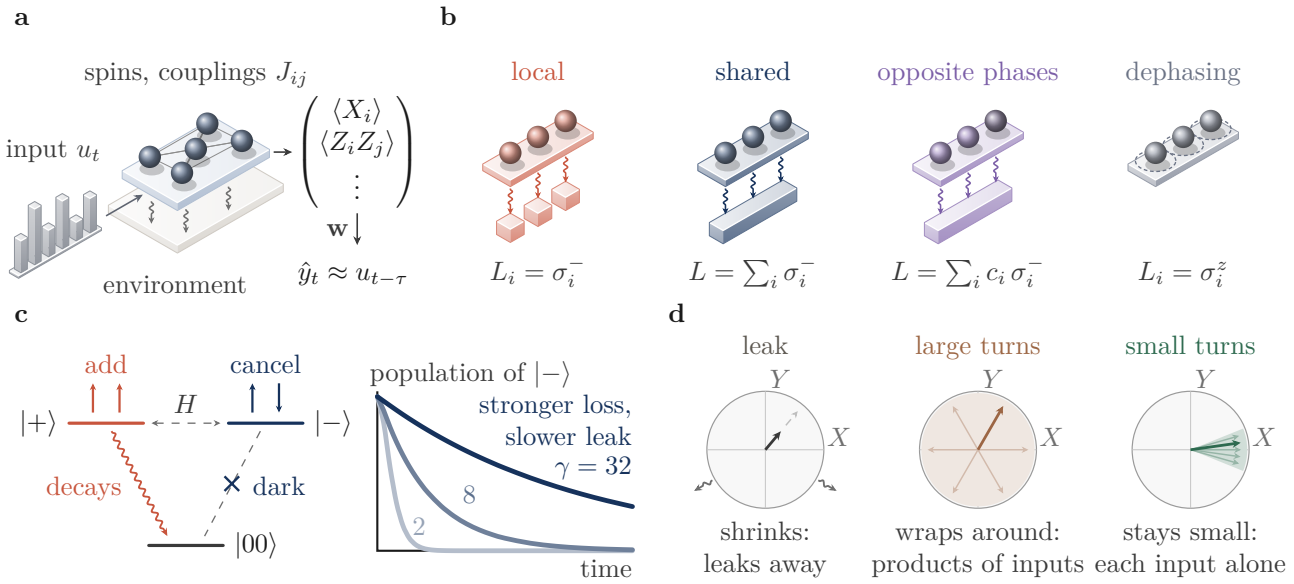


Figure 1: **Engineering how a quantum reservoir leaks. a** Inputs drive interacting spins coupled to an environment; a linear readout recalls past inputs from the spins' expectation values. **b** Four shapes of the environment at equal total coupling, with the baths that take what the spins lose; $c_i = \pm 1$, darker spin: reversed phase. **c** Interference makes dark states: for two spins sharing one channel, the decay amplitudes of $|10\rangle$ and $|01\rangle$ add for $|+\rangle$ and cancel for $|-\rangle$. The Hamiltonian couples the two, and strong loss turns this into a slow leak (inset: population of $|-\rangle$ at $\gamma = 2$, 8 and 32, in units of their coupling). **d** Two ways to forget: a stored input leaks away, and later inputs, each turning the stored state by $\pm\theta$, spread its direction by about $\theta\sqrt{t_{\mathrm{mem}}}$. Small turns keep each input readable on its own; large turns leave only products of inputs.

quantum Zeno effect [29–31]. Second, the input must follow this clock. Later inputs scramble a stored input (in the everyday sense, not many-body information scrambling) into products that a linear readout cannot use. For reservoirs in which a stored signal fades along a single slow direction, we prove that the number of inputs mixed into a memory is exponentially distributed, with mean equal to how hard one input nudges it times how long it lasts. An input that gets gentler as the memory grows therefore more than doubles the linear memory with exact readout; the phases of a shared channel decide whether a uniform input writes gently on its own. Third, the shape decides how large, and hence how measurable, a memory is. Dissipation that cools the spins leaves a state the input can tilt; once measurements are counted, weak local loss keeps about twice the memory of the dark states, which last as long but are more than ten times fainter.

Some ingredients are known: short-term memory and its bounds in echo-state and linear networks [2, 32, 33], the trade of memory against nonlinearity through the input scaling [34, 35], recently across quantum reservoir protocols [36], and the slow leak out of Zeno-protected states [30, 37]; weakening the input as the memory grows is the quantum counterpart of shrinking the input scaling near the edge of stability in leaky-integrator echo-state networks [38]. New is what the jump operators do to memory, through their interference, their phases and the state they leave behind, and the law for how many inputs a memory mixes, which ties the input to the clock. The law is proved for a class of reservoirs (Appendix A), the three-spin limits are certified by interval arithmetic (Appendix C), and the five-spin results are numerical.

2 Same total coupling, different shapes

We use the driven spin reservoir of Ref. [7]. Five spins interact through random couplings J_{ij}, and in each step a random input u between 0 and 1 sets the strength of a transverse field for a time Δt:

$$H(u) = \sum_{i<j} J_{ij} X_i X_j + h \sum_i Z_i + h(1+u) \sum_i X_i. \tag{1}$$

The environment acts through jump operators L_k in a Lindblad master equation [39, 40],

$$\dot{\rho} = -i[H(u), \rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \Big), \tag{2}$$

Table 1: Shapes of the environment at five spins, each with $\operatorname{Tr} K = 80\gamma$, and the number of operators that the dissipator leaves unchanged; $i < j$ for pairwise loss, $i \neq j$ for exchange and $c = (1, 1, 1, -1, -1)$.

shape	jump operators	protected
local	$\sqrt{\gamma}\,\sigma_i^-$	1
shared	$\sqrt{\gamma}\sum_i \sigma_i^-$	100
opposite phases	$\sqrt{\gamma}\sum_i c_i\sigma_i^-$	100
two channels	$\sqrt{\gamma}\sum_{i\leq 3}\sigma_i^-$, $\sqrt{\gamma}\sum_{i>3}\sigma_i^-$	36
pairwise	$\sqrt{\gamma}\,\sigma_i^-\sigma_j^-$	36
exchange	$\sqrt{0.4\gamma}\,\sigma_i^-$, $\sqrt{0.3\gamma}\,\sigma_i^-\sigma_j^+$	1
gain/loss	$\sqrt{\gamma/1.3}\,\sigma_i^-$, $\sqrt{0.3\gamma/1.3}\,\sigma_i^+$	1
dephasing	$\sqrt{\gamma/2}\,\sigma_i^z$	32

with all L_k scaled by one loss rate γ. A linear readout of 45 features, the single-spin and two-spin expectation values, is trained to name the input from τ steps back. Its recall C_τ, between 0 and 1, is the squared correlation between the estimate and the input on held-out data. The memory $\sum_\tau C_\tau$ counts roughly how many past inputs the reservoir can name, in units with a ceiling of 45; the capacity counts products of inputs as well. The memory time t_{mem}, how long a stored signal of any kind survives the leak (Sec. 5.4), is the clock of the reservoir. Secs. 3.1–3.3 use exact expectation values and an unregularized readout, an idealization that credits arbitrarily small signals; Sec. 3.4 counts measurements and decides which clock stays readable.

We call the set of jump operators the shape of the environment and γ its strength. Every shape gets the same total coupling $\operatorname{Tr} K = \sum_k \operatorname{Tr} L_k^\dagger L_k$, which fixes the mean decay rate over all 4^N modes of the reservoir, for every Hamiltonian (Sec. 5.2). It does not fix the slow modes, which can move in opposite directions, so Sec. 3.4 also compares the shapes at a matched clock. Of the eight shapes in Table 1, four carry the story (Fig. 1b): local loss, where every spin loses its excitation on its own; one shared channel, where all spins lose it together with equal phases; the same channel with opposite phases on spins 4 and 5; and dephasing, where every spin loses its phase.

What tells the shapes apart is what the environment cannot reach. Local loss leaves only the ground state alone. A shared channel, with either phases, cannot empty ten dark states and leaves every operator on them, 100 in all, unchanged, while dephasing keeps the 32 populations of the spin basis but erases every coherence.

3 Results

The argument has three steps: which way the clock runs, how hard the input may write, and which clock stays readable once measurements are counted.

3.1 The shape sets the direction of the clock

At weak loss all shapes behave alike; as γ grows they split in opposite directions (Fig. 2). With local loss, recall retreats to ever shorter delays and the memory falls steadily (Fig. 2a,c). With one shared channel the memory first dips and then doubles, and at strong loss inputs fifty steps back are still recalled (Fig. 2b,c). The decay rates show the split (Fig. 2d): without protected operators the leak grows in proportion to γ and the clock speeds up, while with dark states the leak falls as $1/\gamma$ and the clock slows down.

Why does more loss slow a shared channel down? Take two spins that share one channel (Fig. 1c). Their decay amplitudes cancel for the dark state $|-\rangle$, which can leak only after the Hamiltonian turns it into the bright state $|+\rangle$. A strong channel acts like constant checking of whether an excitation has left, and such checking holds the state where the answer is no [29, 41], the quantum Zeno effect: the leak falls as Ω^2/γ, where Ω couples $|+\rangle$ and $|-\rangle$, just as the slow mode of an overdamped oscillator settles more slowly the stronger the friction.

The same holds for any number of spins. At strong loss the reservoir follows its Hamiltonian restricted to the dark states, which turns the stored state but cannot make it forget; forgetting enters only at the next order, when the Hamiltonian briefly lifts a dark state into a bright one that decays [30, 37]. One step is then a rotation plus a leak that falls as $1/\gamma$ (Eq. (4)): a nearly orthogonal map at the edge of stability, the classic recipe for a long memory in a network [32]. The size of the reservoir sets only the strength of the leak (Appendix D). Local loss has no such regime: at strong loss it protects only the ground state, and a single state cannot hold a record.

Two tests show that the dark states hold the memory (Appendix D). A model of the ten dark states alone, with no fitted parameter, reproduces the recall of the full reservoir within a few percent once the loss is moderately strong. Erasing the dark coordinates after every step destroys

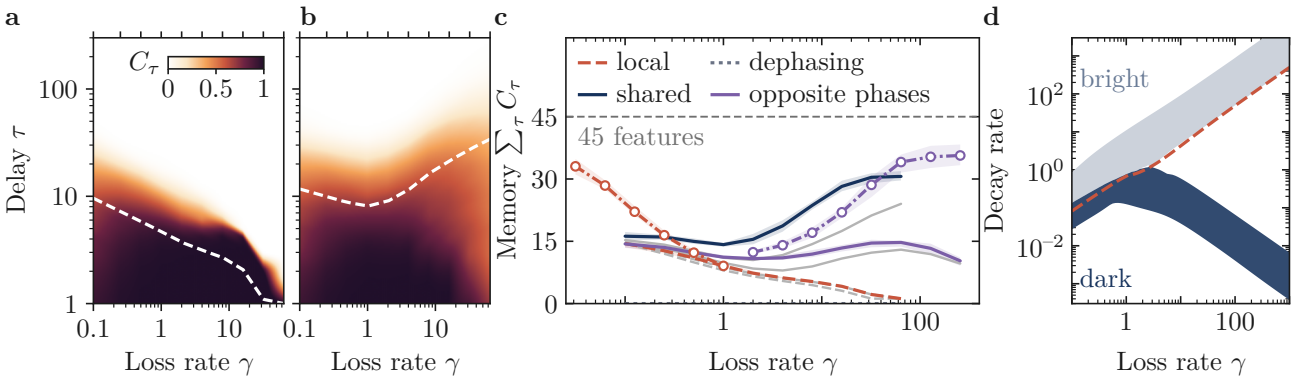


Figure 2: **Same total coupling, opposite clocks. a,b** Recall C_τ of the input τ steps back against the loss rate, with local loss (a) and one shared channel (b); dashed: the delay below which half of the memory lies. **c** Memory at the full input amplitude, for the four shapes that carry the story and, in grey, the others: exchange and gain/loss (dashed) follow local loss, two channels the shared one, and pairwise loss opposite phases; one and two shared channels end at $\gamma = 64$ (Appendix D); circles: gentle writing (Sec. 3.3). **d** Decay rates of one reservoir at the mean input: bright and dark modes of a shared channel; dashed: the slowest mode of local loss.

all memory beyond a few steps, while erasing a rotated copy of the same size leaves most of it.

But lasting is not the same as staying readable. With opposite phases the dark states are just as protected, yet the memory climbs back only to its weak-loss level (Fig. 2c): something besides the leak is erasing it. Dephasing, too, slows the clock of its protected populations, yet stores nothing (Sec. 3.4).

3.2 The price of a long memory

That something is the input itself. Every input nudges the stored state a little, and the nudge depends on the input (Fig. 1d): to an old input, every newer input is noise. Over the memory time the stored direction takes a random walk of small turns $\pm\theta$ and spreads by about $\theta\sqrt{t_{\mathrm{mem}}}$. Once the spread reaches about a radian, the record of an old input survives only multiplied by the inputs that followed it, and a linear readout cannot turn such products back into single inputs. We say that later inputs scramble the stored input.

A single input is thus lost in two ways: it leaks away, and it is scrambled. The time to forget t_{f}, how long one input stays linearly readable, adds both: its rate is the leak plus a scrambling rate that grows as η^2, the squared input amplitude (Sec. 5.3). Because the leak grows as γ with local loss and falls as $1/\gamma$ with dark states, $\eta^2 t_{\mathrm{f}}$ should depend on loss and input only through γ/η^2 with local loss and through $\gamma\eta^2$ with opposite phases. Both collapses hold over a range of 64 in η^2, and they run in opposite directions (Fig. 3a); they test that leak and scrambling add, not which products are stored. With exact readout, then, no loss rate is best on its own: weakening the input moves the best loss rate (Appendix D).

Here the phases of the channel become a design choice. The dark states are annihilated by the collective lowering operator $\sum_i c_i \sigma_i^-$, and an input field $\sum_i c_i X_i$ with the same phases is that operator plus its adjoint, so it cannot couple two dark states: it writes only through brief excursions into bright states, and its effect falls as $1/\gamma$. An input pattern across the phases writes into the dark states directly, and its effect falls only slowly (Fig. 3b). The uniform input of Eq. (1) is of the first kind with equal phases, so its time to forget keeps growing in proportion to γ, and of the second with opposite phases, so its time to forget levels off near 56 steps. The choice is fine-tuned: a phase error δ couples dark states directly at order δ, and beyond $\gamma \sim 1/\delta$ the equal-phase channel behaves like the one with opposite phases.

How many inputs does a scrambled memory mix? We call the number of distinct inputs that a stored function involves its order: u_{t-3} alone has order one, $u_{t-3}u_{t-5}$ order two, which splits the information processing capacity [34, 42] by inputs. The simplest reservoir with a long memory is reset to a fixed state with a small probability p in each step and otherwise turned by an input-dependent unitary; a stored signal then lasts about $t_{\mathrm{mem}} = 1/(2p)$ steps. How hard one input nudges what is stored is measured by its mixing fraction s, between 0 and 1: the part of a stored signal that one input turns away from the average turn

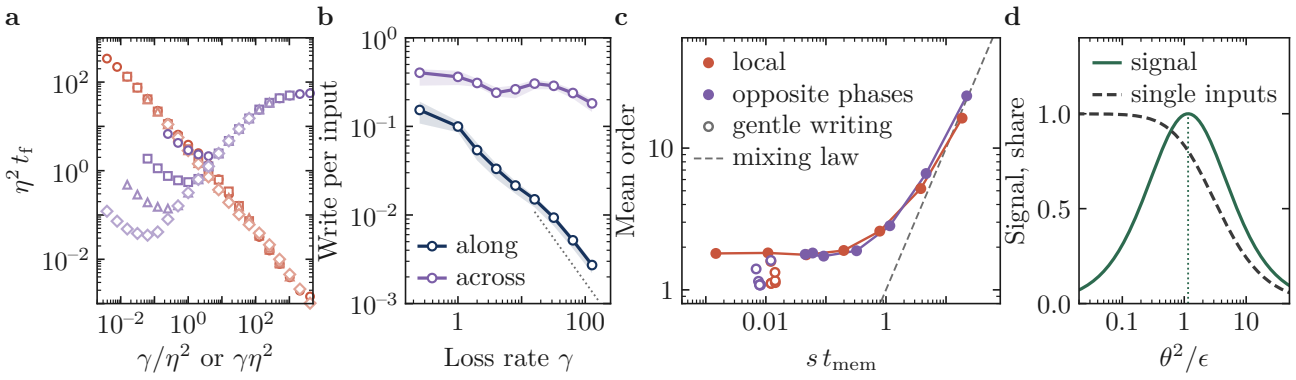


Figure 3: **The price of a long memory, and how the input pays it. a** Time to forget t_f at input amplitudes $\eta = 1$, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ (circles, squares, triangles, diamonds), times η^2, against γ/η^2 for local loss and $\gamma\eta^2$ for opposite phases. **b** Write per input into the dark states of the equal-phase channel, for the input along the phases (uniform) and across them (cosine pattern); dotted: $1/\gamma$. **c** Mean order of the capacity against the mixing fraction times the memory time, at the full amplitude (filled; opposite phases from $\gamma = 1$ to 4096, local loss from 4 to $1/1024$) and with gentle writing (open); dashed: the mixing law. **d** One qubit in the limit $\epsilon \to 0$: stored signal and share of single past inputs against the squared turn per memory time; dotted: the largest signal.

(Appendix A). For small turns it grows as the square of the turn.

> **Theorem 1** (Mixing law, informal; precise form in Appendix A)**.** Let a reservoir be reset with probability p per step and otherwise turned by input-dependent unitaries that mix the stored state, leave a single slow direction of the stored covariance, and write a signal (conditions in Appendix A). Pick a stored function with probability equal to its share of the capacity, and let D be its order. As $p \to 0$, D/t_{mem} becomes exponentially distributed with mean s, and
>
> $$\mathbb{E}D = s\,t_{mem} + O(1).$$

In words, a memory mixes on average $s\,t_{mem}$ inputs: how hard one input nudges it times how long it lasts. The reason is a race: in each step a fraction s of the stored signal picks up one more input, while a fraction $1/t_{mem}$ of it is lost. The law holds for physical channels without a reset as well, with the leak in place of the reset (Appendix A.5). What it needs is a single slow direction of the stored covariance, not a single slow mode.

In the five-spin reservoir the moments give the order distribution exactly (Eq. (5)). One input turns the stored state only slightly, $s \approx 0.02$ at the full amplitude. As the memory grows, the mean order with opposite phases rises from 1.8 at $\gamma = 1$ to 23 at $\gamma = 4096$, as the mixing law predicts: it exceeds $s\,t_{mem}$ by 1.3 to 2.0 throughout (Fig. 3c), and gentle writing (Sec. 3.3) keeps it near one. Mixing is not only a cost: products of inputs are the nonlinear memory that tasks such as parity need [34, 35], and the equal-phase channel, whose uniform input writes gently by itself, keeps about half as much of it as local loss (Appendix D). At a fixed input strength, though, a longer memory is a more scrambled one.

3.3 The input must follow the clock

How much gentler must the input get? One qubit shows the trade-off in closed form (Fig. 3d, Appendix B). It relaxes towards $+X$ at a small rate ϵ, each input turns it about Z by $\pm\theta$, and it is read along Y. Everything depends on θ^2/ϵ, the squared turn per memory time. When it is large, later inputs scramble earlier ones; when it is small, each input writes too little. The signal is largest in between, at $\theta^2/\epsilon = 2/\sqrt{3}$, and a turn that falls exactly as $\epsilon^{1/2}$ is the only power law that keeps it finite. We call an input amplitude that follows the clock in this way, falling as one over the square root of the memory time, gentle writing: it keeps $s\,t_{mem}$ fixed as the memory grows.

The five spins follow the same rule in both directions of the clock: the amplitude falls as $1/\sqrt{\gamma}$ where the clock slows with stronger loss, and as $\sqrt{\gamma}$ where it slows with weaker loss (Fig. 2c, circles). With exact readout this more than doubles the linear memory: from 14 to 33 units with local loss at $\gamma = \frac{1}{32}$, and from the best 14.7 at the full amplitude to 36 with opposite phases (Table 2).

Table 2: Linear memory $\sum_\tau C_\tau$ of the five-spin reservoir in units (ceiling 45), mean of eight reservoirs; exact readout, or a budget of 10^9 preparations times replayed steps per output step.

	local loss	opposite phases	equal phases
tuned loss, $\eta = 1$	14.5	14.7	31[a]
weak input[b]	16	25	–
gentle writing	33 ($\gamma = \frac{1}{32}$)	36 ($\gamma = 256$)	–[c]
10^9, $\eta = 1$[d]	15 ($\gamma \approx \frac{1}{16}$)	12 ($\gamma \approx 64$)	12 ($\gamma = 1$)[e]
10^9, gentle writing	29 ($\gamma = \frac{1}{64}$)	15 ($\gamma \approx 64$)	–[c]

[a]At strong loss. [b]At $\gamma = 1$ and $\eta = \frac{1}{32}$, two reservoirs. [c]The uniform input already writes gently in this channel. [d]Upper bounds: replaying τ_{90} charges too little at the full amplitude. [e]Smallest loss rate sampled; the memory falls as the loss grows.

It needs both knobs: weakening the input alone, or tuning the loss alone, falls short. Only the equal-phase channel comes close at the full amplitude, because its uniform input already writes gently.

The gain is a reallocation, not free capacity. With exact readout all capacities of a readout add up to its rank, 45 here [34], so raising the linear memory from 14 to 33 units leaves at most 12 instead of 31 for products of inputs. With opposite phases at $\gamma = 256$, single past inputs carry 89% of the capacity with gentle writing and 28% at the full amplitude. Dark states are therefore not what makes a memory long. They keep it long while the reservoir dissipates strongly, and they reverse the direction of the clock.

Three spins with one shared channel of opposite phases show both regimes, and there we prove them (Appendix C): at a fixed amplitude the order grows in proportion to γ and single inputs lose their share, while with an amplitude $\eta_0/\sqrt{\gamma}$ they keep a finite share and their recall reaches ever further back.

The memory is written into coherences between dark energy levels; coherence is known to shape the performance of spin reservoirs [43]. Erasing these coherences after every input leaves the slow population dynamics alike but collapses the stored signal as $1/\gamma^2$, because a new input first turns a population into a coherence (Appendix C). A gentler input writes a fainter record. With exact readout this costs nothing; with real measurements, faintness has a price.

3.4 What survives measurement: cool, don't randomize

Exact expectation values stand for averages over infinitely many runs, and they flatter a long memory. A memory that follows the clock grows longer, not larger: it cannot exceed the 45 features, so each delay is recalled more faintly. With local loss at $\gamma = 64$ the previous input still moves the state by about 10^{-9}, enough for a perfect recall with exact readout, but at 10^8 preparations per step its recall drops to 0.002.

Finite sampling limits what any physical reservoir resolves [44], so we estimate every feature from n preparations per measurement setting and step, with all spins read along X, Y or Z, and choose the regularization of the readout on held-out data (Sec. 5.5). Each preparation must also replay the input history that the memory still holds [13], and this history grows with the clock. A fair comparison therefore fixes the budget, preparations times replayed steps per output step: three settings, each with n preparations that replay the range τ_{90} holding 90% of the exact recall. At the full amplitude this charges too little, so those curves are upper bounds.

At a fixed budget the loss rate has an optimum again, and the budget sets it (Fig. 4a). At 10^9 preparations times replayed steps, weak local loss with gentle writing keeps the most, 29 units, against 15 for opposite phases at their optimum near $\gamma = 64$ (Table 2). Gentle writing pays once each setting is prepared about 10^5 times per step with local loss and 2×10^6 times with opposite phases. The equal-phase channel, whose exact memory grows with the loss, now loses memory as the loss grows, and the dark-state memory, as long as that of weak local loss, is more than ten times fainter.

What sets the size of the signal appears to be the state that the dissipation leaves behind. An input acts on the stationary state ρ_* through the commutator with its Hamiltonian, and only the part of this change that lies on the slow modes is remembered; we call its size the write onto the slow modes. At one matched clock and input amplitude, the feature variance of the seven shapes that write follows this write over four orders of magnitude (Fig. 4b). Dephasing, being unital, leaves a fully mixed stationary state that the input cannot tilt, and it writes and stores nothing. Shared channels leave a polarized state

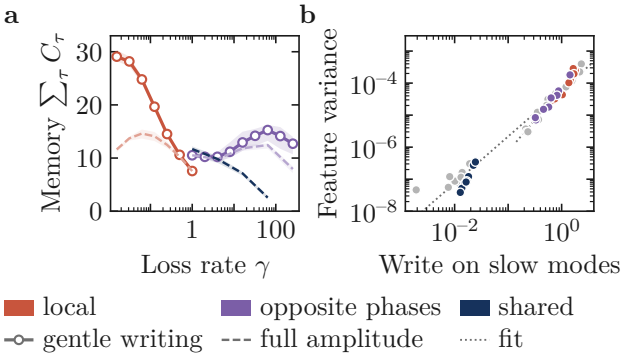


Figure 4: **What survives measurement, and why.** **a** Memory at 10^9 preparations times replayed steps per output step, against the loss rate, with gentle writing or at the full amplitude. **b** Feature variance against the write onto the slow modes, $\|\Pi_{\text{slow}}[H_{\text{in}}, \rho_*]\|$, for the seven shapes that write, at a matched clock and input amplitude, eight reservoirs each, coloured as in Fig. 2c; dotted: power laws fitted without and with dark states, with exponents 2.0, as linear accumulation of the write predicts, and 1.5, which the write alone does not explain.

that the uniform input tilts only through bright states, and they store almost nothing; local and pairwise loss cool the spins into a state that the input tilts directly, and they store the most. With earlier findings that non-unital damping helps and unital noise hurts [8, 45], this gives a heuristic, shown here for one clock, one class of Hamiltonians and a uniform input: a reservoir should forget by cooling, towards a low-entropy state, not by randomizing.

4 Discussion

The shape of the dissipation matters as much as its rate, and three design rules follow for the reservoirs we studied:

(1) let the jump operators set the direction of the clock: separate channels slow it with weaker loss, a shared channel with dark states slows it with stronger loss, and its phases decide whether the input reaches the dark states;

(2) let the input follow the clock, weakening it as the memory grows so that the mixing fraction of one input times the memory time stays of order one;

(3) as a heuristic, forget by cooling, not by randomizing: prefer dissipation that leaves a polarized state the input can tilt.

A damping sweet spot found at a fixed input scaling [7] is therefore not a property of the dissipation alone: it moves as the input weakens, and with finite sampling the measurement budget sets it. The right shape need not be exotic: with a uniform input, weak local loss keeps the most memory at every budget we tried.

Dark states pay when the reservoir must dissipate strongly, as in waveguides, where collective decay comes with the coupling [46, 47] and the positions of the spins set the phases of the channel: spins half a wavelength apart couple with opposite signs [47]. Ordinary noise limits their reach: local loss or dephasing at a small rate adds to the leak out of the dark states, which the collective loss suppresses only as $1/\gamma$, so a memory of T steps needs collective loss about $4T^2$ times stronger than local loss (Appendix D). A dark state of four transmons in a waveguide lives more than a hundred times longer than a single qubit decaying into the guide [46]; read as such a ratio of rates, this allows a memory of about five steps, and the fifty steps of Fig. 2b would need a ratio near 10^4.

Parts of this story have classical cousins: two oscillators sharing one damper keep an undamped mode, and the $1/\gamma$ leak is the slow mode of an overdamped oscillator. The quantum structure adds three things: the protected set is a space of operators, 100 of them for ten dark states; the memory is written into coherences between dark energy levels; and the phases of the jump operators and the input decide whether the input reaches the protected states at all.

Our results hold for one class of Hamiltonians, eight reservoirs and a uniform input, whose symmetry decides which channels write gently by themselves; with the input on a single spin the collapses persist (Appendix D). We study memory and capacities, which set what a linear readout can reproduce [34], not a particular task. Jumps beside a transverse drive assume a bath spectrum that is flat over the relevant frequencies [48]; an open superconducting waveguide provides one up to $\gamma \approx 64$ for couplings of order a megahertz [47], and shared resonators, waveguides and structured couplers can build such channels [47–50]. Concurrent work finds that decoherence-free modes break the echo-state property and that responses below the regularization scale do not count [51]; our dark states leak, and Sec. 3.4 shows the second effect. Another finds a balance between memory

and nonlinearity across reservoir protocols [36], which the mixing law makes quantitative for slow memories. Truncated replay [13] or weak measurements [10] would cut the cost of sampling; how a slow clock combines with non-Markovian memory [14] is open.

How a quantum system leaks decides not only how fast it forgets, but what it keeps and in what form.

5 Methods

5.1 Reservoir, readout and memory

The Hamiltonian of Eq. (1) uses $h = 1/2$ and couplings J_{ij} drawn uniformly from $[-1, 1]$, so the transverse field runs from h to $2h$. Each input u is drawn uniformly from $[0, 1]$ and held for a step of length $\Delta t = 1/2$. We integrate Eq. (2) in the 1024-dimensional basis of Pauli operators and evaluate the step map $E(u) = e^{\Delta t \mathcal{L}(u)}$ by Chebyshev interpolation in u at ten nodes. Each reservoir draws its couplings and its input stream from one seed, and every shape uses the same eight seeds. After a washout of $\max(800, 60\gamma)$ steps we record the 45 features, the expectation values of X_i, Y_i, Z_i and of X_iX_j, Y_iY_j, Z_iZ_j, for 2000 training, 400 validation and 1600 test inputs. The readout is a least-squares fit with an intercept on the training and validation inputs, and C_τ is the squared correlation of its estimate with $u_{t-\tau}$ on the test inputs. The memory sums C_τ over delays 1 to 300; a delay without any recall reads about $1/1600$ on the test inputs, so the sum has a floor of about 0.2 units. Figures show the mean of the eight reservoirs, with bands from the 16th to the 84th percentile; recall maps and curves are smoothed over neighbouring delays, and the maps interpolate between the computed loss rates, which ticks on their upper edge mark (Appendix D).

5.2 Shapes and equal coupling

Table 1 lists the jump operators. Their weights give every shape the same total coupling $\operatorname{Tr} K = N2^{N-1}\gamma = 80\gamma$; any positive mixture in exchange and gain/loss leaves a single stationary state, as local loss does. The total coupling fixes the mean decay rate. For a traceless jump L the superoperator trace of its dissipator is $-2^N \operatorname{Tr} L^\dagger L$, while the commutator with any Hamiltonian is traceless, so the 4^N eigenvalues λ of the generator obey

$$\frac{1}{4^N} \sum \operatorname{Re} \lambda = -\frac{\operatorname{Tr} K}{2^N} \tag{3}$$

for every shape and every Hamiltonian. The protected operators form the kernel of the unit-rate dissipator.

5.3 Slow modes, time to forget and the strong-loss generator

The input-averaged step $\bar{E} = \mathbb{E}_u E(u)$ is computed by six-point Gauss–Legendre quadrature. Its eigenvalues λ_j give decay rates $\kappa_j = -\ln|\lambda_j|/\Delta t$. If the dissipator protects n operators, the slow modes are the $n-1$ slowest decaying ones besides the stationary state, and the time to forget is $t_\mathrm{f} = 1/(\Delta t\, \bar{\kappa})$, with $\bar{\kappa}$ their mean rate; without protected operators we use the slowest mode. The mean rate adds the leak and the scrambling exactly (Appendix D). Figure 2d shows instead the decay rates $-\operatorname{Re}\lambda$ of the generator at the mean input, $u = 1/2$; with a shared channel we call the $n-1$ slowest of them dark and all faster ones bright.

Let $\mathcal{D}$ be the unit-rate dissipator, $\mathcal{P} = \lim_{t\to\infty} e^{t\mathcal{D}}$ the projector onto its kernel and $\mathcal{S} = (\mathcal{D} + \mathcal{P})^{-1} - \mathcal{P}$ its inverse on the decaying operators. At strong loss, eliminating the decaying operators to second order [30, 37] gives the generator of the protected coordinates,

$$\mathcal{G}(u) = \mathcal{P}\mathcal{H}(u)\mathcal{P} - \frac{1}{\gamma}\mathcal{P}\mathcal{H}(u)\mathcal{S}\mathcal{H}(u)\mathcal{P}, \tag{4}$$

with $\mathcal{H} = -i[H, \cdot]$. The first term is the Hamiltonian restricted to the protected operators, which turns them; the second is the leak through brief excursions into bright states.

For gentle writing we replace $h(1+u)$ in Eq. (1) by $h[\frac{3}{2} + \eta(u - \frac{1}{2})]$, which keeps the mean field and scales the input amplitude: $\eta = \min(1, \gamma^{-1/2})$ for opposite phases and $\eta = \min(1, \gamma^{1/2})$ for local loss.

5.4 Order of a memory

For independent inputs, every function of the past inputs with finite variance splits uniquely into orthogonal parts, each of which depends on a fixed finite set of inputs; the order of a part is the size of its set. For signs $u_t = \pm 1$ the parts are the products $\prod_{j\in J} u_{t-j}$; for continuous

inputs one uses orthogonal polynomials in each input. Quantum channels act affinely on traceless coordinates [52, 53], $r_t = A(u_t)r_{t-1} + b(u_t)$. Let $\bar{A} = \mathbb{E}A$ be the mean step, m the stationary mean and $\xi = A(u)m + b(u) - m$ the innovation, the change that one input writes, with covariance V. With $\mathcal{B}X = \bar{A}X\bar{A}^{\mathsf{T}}$, $\mathcal{T}X = \mathbb{E}[AXA^{\mathsf{T}}]$ and $\mathcal{K} = \mathcal{T} - \mathcal{B}$, the covariance G_k carried by products of order k obeys

$$G_1 = (1-\mathcal{B})^{-1}V, \qquad G_{k+1} = (1-\mathcal{B})^{-1}\mathcal{K}G_k. \quad (5)$$

The innovation adds the current input alone; the mean step carries a stored product along without adding an input; and the fluctuating part $\mathcal{K}$ of the step multiplies it by the current input, which raises its order by one. The G_k add up to the stationary covariance Σ. A readout of features $y = Mr$ explains the share $\mathrm{Tr}[(M\Sigma M^{\mathsf{T}})^{+}MG_kM^{\mathsf{T}}]$ of order k, and these shares add up to the rank of the readout [34]. For one observable with coordinates a the share of order k is $a^{\mathsf{T}}G_ka/a^{\mathsf{T}}\Sigma a$, and the share of single past inputs is $a^{\mathsf{T}}\bar{A}G_1\bar{A}^{\mathsf{T}}a/a^{\mathsf{T}}\Sigma a$.

The memory time and the mixing fraction come from the slowest mode of the covariance step $\mathcal{T}$: its eigenvalue is $1 - 1/t_{\mathrm{mem}}$, and with its right eigenvector Q and left eigenvector ℓ the mixing fraction is $s = \ell(\mathcal{K}Q)/\ell(Q)$, as in Appendix A.5. The memory time thus measures how long the variance of a stored signal survives the leak under the random inputs. For the five-spin reservoir we evaluate Eq. (5) on the 1023 traceless Pauli coordinates (Appendix D).

5.5 Finite sampling

For each step we form the outcome distributions of the three settings from the exact state and draw n preparations from each. The readout is fitted as above, with a ridge chosen from fourteen values between 0 and 10^4 on the validation inputs. For a budget B of preparations times replayed steps per output step we measure the memory at $n = 10^5$ to $10^{8.5}$ preparations in half decades, with the range τ_{90} that holds 90% of the exact recall, and interpolate in $\log n$ to $n = B/(3\tau_{90})$, on eight reservoirs; for the lines compared in Sec. 3.4 we add $n = 10^3$ to $10^{4.5}$. The matched clock of Fig. 4b is a leak time $t_{\mathrm{leak}} = 1/(\Delta t\,\bar{\kappa}_0)$ of 32 steps, where $\bar{\kappa}_0$ is the mean decay rate of the slow modes of the generator at the mean input (Appendix D). It measures how long the amplitude of these modes survives without the random inputs, while t_{mem} measures how long the variance of a stored signal survives with them; both are set by the leak.

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Data availability

The code, the data behind every figure and the interval certificates of the three-spin limits are available at https://github.com/eybmits/stronger-dissipation-longer-memory; the certificates are also attached to this PDF.

Author contributions

M.B. contributed to conception, methodology, software, analysis, visualization and writing. I.F. and J.W. contributed to methodology, interpretation and editing. C.L.-P. and J.S. contributed to supervision and conceptual development. Two AI assistants were used. OpenAI Codex assisted with drafting, literature checks, derivations, code, numerical experiments and the verification software, including the certificate programs of Appendix C and a second, separate finite-rate implementation that shares no routine with them. Anthropic's Claude Code assisted with drafting and editing the text, the figures, the numerical checks of Secs. 3.3 and 3.4, and the runs of the clock and sampling tests. All AI-assisted material was reviewed and verified by the authors, who take full responsibility for the scientific content and final manuscript.

A The mixing law

This appendix proves the mixing law of Sec. 3.2: in a reservoir that forgets slowly, a stored function involves on average as many inputs as the mixing fraction of one input times the memory time. The last part extends the law to physical channels without a reset, which Appendix C needs.

A.1 Setting and statement

We describe the reservoir by the n traceless coordinates r of its state in an orthonormal basis of operators, the Bloch vector for a qubit. A unitary acts on r as a rotation. In each step the reservoir is reset to a fixed state with a small probability p and is otherwise rotated by $O(u)$:

$$r_t = (1-p)\,O(u_t)\,r_{t-1} + p\,b. \tag{A1}$$

Here b represents the reset state, and the inputs u_t are independent and identically distributed. A stored signal loses a fraction $2p$ of its variance per step, so it lasts about $t_{\mathrm{mem}} = 1/(2p)$ steps. Three conditions make the law hold:

(1) *The average step forgets.* The average rotation $A = \mathbb{E}O$ shrinks every vector: its spectral radius is below one.

(2) *The inputs scramble.* The only covariance matrices that the random rotations leave unchanged on average, $\mathbb{E}[OXO^{\mathsf{T}}] = X$, are multiples of the identity.

(3) *The input writes.* One input changes the state: $\mathbb{E}\|(O-A)\mu\|^2 > 0$ with $\mu = (1-A)^{-1}b$.

The first condition means that only products of inputs can be stored for long, the second that a stored signal spreads evenly over all directions, and the third that there is something to store. The mixing fraction

$$s = 1 - \frac{\|A\|_F^2}{n} \tag{A2}$$

says how hard one input nudges the stored state. For a unit vector x of random orientation, $\mathbb{E}\|(O-A)x\|^2 = s$: one input turns a fraction s of a stored signal away from the average rotation. The fraction lies between 0 and 1, and the first condition makes it positive. If the inputs $u = \pm 1$ rotate by $O = e^{u\theta G}$ with a fixed generator G, whose eigenvalues are $i\lambda_1, \dots, i\lambda_n$, the average rotation is $\cosh \theta G$ and $s = \frac{1}{n}\sum_j \sin^2 \theta\lambda_j$, which grows as θ^2 for small turns.

Mixing law, precise form. Assume conditions (1)–(3) and take any fixed linear readout. Pick a stored function with probability equal to its share of the readout's capacity, and let D be its order. As $p \to 0$, the ratio D/t_{mem} becomes exponentially distributed with mean s, and

$$\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1).$$

Equivalently, products of more than $x\,t_{\mathrm{mem}}$ inputs carry the share $e^{-x/s}$ of the capacity.

A.2 The idea: a race

A stored signal is spread evenly over all directions. In each step the current input turns a fraction s of it away from the average rotation, which adds one input to that part, and the reset removes a fraction $1/t_{\mathrm{mem}}$. On average $1/s$ steps pass before the next input joins a product, and meanwhile it loses a fraction $1/(s\,t_{\mathrm{mem}})$ of its weight, so products of more than K inputs carry about

$$\Big(1 - \frac{1}{s\,t_{\mathrm{mem}}}\Big)^K \approx e^{-K/(s\,t_{\mathrm{mem}})},$$

an exponential law with mean $s\,t_{\mathrm{mem}}$. The proof makes this bookkeeping exact.

A.3 Proof

The proof works with covariance matrices. Methods splits the stationary covariance Σ of the state into the parts G_k carried by products of k inputs and gives their recursion, Eq. (5): the mean step $\mathcal{B}$ carries a stored product along, and the fluctuating part $\mathcal{K}$ of the step adds one input to it. In the reset model the mean step is $\mathcal{B}X = (1-p)^2 AXA^{\mathsf{T}}$, the full step $\mathcal{T}X = (1-p)^2\,\mathbb{E}[OXO^{\mathsf{T}}]$, and $\mathcal{K} = \mathcal{T} - \mathcal{B}$. We write $\mathcal{B}_0$, $\mathcal{T}_0$ and $\mathcal{K}_0$ for the same maps at $p = 0$.

Step 1: the stored signal is small and spread evenly. The stationary mean is $m = p\,[1-(1-p)A]^{-1}b = p\mu + O(p^2)$, so one input writes the change $(1-p)(O-A)m$, of size p. Its covariance V has the trace $p^2 v + O(p^3)$ with $v = \mathbb{E}\|(O-A)\mu\|^2$, which is positive by condition (3). The random rotations preserve the trace of a covariance and, by condition (2), leave only the identity unchanged; since they cannot enlarge a covariance, the identity is a simple fixed point with the projector $X \mapsto (\operatorname{Tr} X/n)\,1$. On every other direction $(1-\mathcal{T})^{-1}$ stays bounded as $p \to 0$, while on the identity it equals $1/[1-(1-p)^2] \approx 1/(2p)$. The written covariance therefore piles up along the identity:

$$\Sigma = (1-\mathcal{T})^{-1}V = \frac{pv}{2n}\,1 + O(p^2). \tag{A3}$$

Step 2: few inputs carry almost nothing. The mean step forgets by condition (1), so $(1-\mathcal{B})^{-1}$ stays bounded as $p \to 0$. The recursion then makes $G_1 = (1-\mathcal{B})^{-1}V$ of order p^2, and the same holds for every G_k with fixed k. Measured against Σ, which is of order p and has full rank, any fixed set of orders carries a share of order p of the capacity of any readout, although the shares of all orders add up to its rank. A long memory consists of products of many inputs.

Step 3: counting the inputs. Call $\mathcal{F} = (1-\mathcal{B})^{-1}\mathcal{K}$ the order transfer, so that $G_{k+1} = \mathcal{F}G_k$. At $p = 0$ it leaves the identity unchanged, because $\mathcal{K}_0 1 = 1 - AA^{\mathsf{T}} = (1-\mathcal{B}_0)1$. Its matching left eigenvector is the weight

$$\ell(X) = \mathrm{Tr}[(1 - A^{\mathsf{T}}A)X], \qquad \ell(1) = ns, \quad \text{(A4)}$$

since $\ell(\mathcal{F}_0 X) = \mathrm{Tr}\,\mathcal{K}_0 X = \ell(X)$ when the rotations preserve the trace. For small p the eigenvalue one moves to

$$1 - \frac{2}{s}\,p + O(p^2)\,: \qquad \text{(A5)}$$

each step up in order costs a fraction $2p/s = 1/(s\,t_{\mathrm{mem}})$ of the weight, as in the race. To see this, differentiate $(1-\mathcal{B})\mathcal{F} = \mathcal{K}$ at $p = 0$ and apply it to the identity, which gives $(1-\mathcal{B}_0)\mathcal{F}_0' 1 = -2\cdot 1$. Its trace is the weight $\ell(\mathcal{F}_0' 1) = -2n$, and dividing by $\ell(1) = ns$ gives the rate $2/s$.

To turn the eigenvalue into a distribution, weigh order k with $e^{-\lambda p k}$. Summing the recursion gives $\sum_k z^k G_k = z(1 - z\mathcal{F})^{-1}G_1$, and as $z = e^{-\lambda p}$ approaches one, only the eigenvalue of Eq. (A5) makes the inverse large. With $\ell(G_1) = p^2 v + O(p^3)$ and $\beta = 2/s$,

$$\sum_{k\geq 1} e^{-\lambda p k} G_k = \frac{pv}{2n}\,\frac{\beta}{\lambda+\beta}\,1 + O(p^2). \qquad \text{(A6)}$$

Divided by Eq. (A3), this is $\beta/(\lambda+\beta)$, the Laplace transform of an exponential distribution of pD with rate β. So $D/t_{\mathrm{mem}} = 2pD$ is exponential with mean s. Because both covariances are multiples of the identity to leading order, the same holds for the share of the capacity of any readout. Finally, $\sum_k k G_k = (1-\mathcal{F})^{-1}\Sigma = \Sigma/(\beta p) + O(p)$ gives $\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1)$.

Two technical points. The eigenvalue one of $\mathcal{F}_0$ is simple: a fixed point of $\mathcal{F}_0$ satisfies $\mathcal{T}_0 X = X$ and is therefore a multiple of the identity, and $\ell(1) > 0$ rules out a Jordan chain. For $p > 0$ every eigenvalue of $\mathcal{F}$ lies inside the unit circle, by a Perron–Frobenius argument: $\mathcal{F}$ maps positive matrices to positive matrices, and so does $(1-\mathcal{F})^{-1} = 1 + (1-\mathcal{T})^{-1}\mathcal{K}$. The eigenvalues of $\mathcal{F}_0$ therefore lie in the closed unit disk, only the simple one equals one, and only the eigenvalue of Eq. (A5) makes the sum diverge.

A.4 When the law needs one slow direction

Condition (2) matters. Let a classical flag, set by the reset state and never changed, select one of two qubit rotations about Z, by $\pm\pi/6$ or by $\pm\pi/3$, and let the reset state put the flag in either value with probability $\frac{1}{2}$ and the qubit along $+X$ in both. Each sector obeys the law with its own mixing fraction, $1/4$ and $3/4$, and a single Y readout sees their mixture, weighted by the variance each sector stores. The gentler sector stores more, so as $p \to 0$ products of more than $x\,t_{\mathrm{mem}}$ inputs carry the share $w\,e^{-4x} + (1-w)\,e^{-4x/3}$ with $w = 0.82$, which exact moments reproduce. One observable then shows two order scales.

A.5 Physical channels without a reset

A physical channel does not reset to a prescribed state, but the proof used the reset only through three facts: one covariance direction is slow, it leaks at a small rate, and the mean step forgets everything else. The same three steps therefore prove the law for general channels. Appendix C uses this version; readers interested in the reset model can skip it.

Setting. One step is $r_t = A_u(\epsilon)r_{t-1} + b_u(\epsilon)$ on the coordinates that the inputs can reach, with a small leak ϵ, and $\mathcal{B}$, $\mathcal{T}$ and $\mathcal{K}$ are formed from A_u as before. We assume:

(i) A_u and b_u depend smoothly on ϵ in mean square, and b_u vanishes at $\epsilon = 0$;

(ii) the mean step forgets at $\epsilon = 0$: $\mathbb{E}A_u(0)$ has spectral radius below one;

(iii) the channel is mean-square stable for small $\epsilon > 0$;

(iv) one covariance direction is slow: $\mathcal{T}$ has a simple eigenvalue $1 - \kappa\epsilon + O(\epsilon^2)$ with $\kappa > 0$ and no other eigenvalue near one; at $\epsilon = 0$ its right eigenvector is a positive matrix Q,

and its left eigenvector is a weight ℓ that is nonnegative on positive matrices, with $\ell(Q) = 1$;

(v) the input writes along it: the change written by one input, of order ϵ, has covariance $\epsilon^2 V_* + O(\epsilon^3)$ with $\ell(V_*) > 0$.

The reset model is the case $\epsilon = p$, $\kappa = 2$, $Q = 1$ and $\ell = \mathrm{Tr}\,/n$. Condition (iv) concerns the covariance that the random steps carry, not the number of slow modes: at a fixed amplitude the inputs turn the 99 slow dark modes of a shared channel with opposite phases differently in each step and spread a stored signal over all of them, so that one covariance direction stays slow. With several slow directions, each mixes at its own rate, as in the example above.

Mixing law without a reset. Under (i)–(v) the mixing fraction $s = \ell(\mathcal{K}_0 Q)$ is positive and at most one, and ϵD becomes exponentially distributed with rate κ/s. The mean order is $s/(\kappa\epsilon) + O(1)$: the mixing fraction times the memory time $1/(\kappa\epsilon)$.

Proof. Steps 1 and 2 carry over. The stationary mean is of order ϵ by (i) and (ii), and the slow eigenvalue piles the written covariance up along Q,

$$\Sigma = \epsilon \frac{\ell(V_*)}{\kappa} Q + O(\epsilon^2),$$

and every fixed order carries a covariance of order ϵ^2. For the memory of a readout with several features we also need Q to have full rank on what the readout measures; for one observable this holds trivially. In step 3, $\mathcal{F}_0$ has the same fixed points as $\mathcal{T}_0$, so Q spans them, and $\ell \circ (1 - \mathcal{B}_0)$ is its left eigenvector, with value $\ell(\mathcal{K}_0 Q) = s$ on Q. Differentiating $(1 - \mathcal{B})\mathcal{F} = \mathcal{K}$ at $\epsilon = 0$ and applying it to Q gives $(1 - \mathcal{B}_0)\mathcal{F}_0' Q = \mathcal{T}_0' Q$, whose weight $\ell(\mathcal{T}_0' Q) = -\kappa$ is the rate of the slow eigenvalue of $\mathcal{T}$. The order eigenvalue is therefore $1 - (\kappa/s)\epsilon + O(\epsilon^2)$, and the Laplace transform follows as before. It remains to show that s is positive, which also makes the eigenvalue simple. Because $\mathcal{K}_0$ maps positive matrices to positive matrices and ℓ is nonnegative on them, $\ell(\mathcal{B}_0 Y) \le \ell(\mathcal{T}_0 Y) = \ell(Y)$ for every positive Y, so $0 \le \ell(\mathcal{B}_0^j \mathcal{K}_0 Q) \le s \le 1$ for every j. If s vanished, the series $\mathcal{F}_0 = \sum_j \mathcal{B}_0^j \mathcal{K}_0$ would give $\ell(\mathcal{F}_0 Q) = 0$, which contradicts $\mathcal{F}_0 Q = Q$ and $\ell(Q) = 1$.

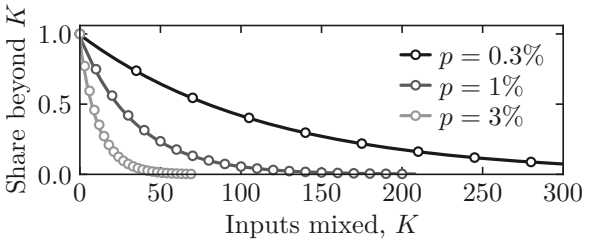


Figure 5: **The mixing law.** Share of the capacity carried by products of more than K inputs, for one qubit reservoir with a reset to a basis state and a single readout, at reset probabilities p per step. Lines: exact moments; circles: the law $e^{-K/(s\,t_{\mathrm{mem}})}$.

A.6 Numerical check

Figure 5 compares the law with exact moments. Beyond the reservoir shown, we tested 24 reservoirs and readouts: in dimensions two, three and four, two families of three random unitaries applied with probabilities 0.2, 0.3 and 0.5, each with two reset states (a basis state and a random pure state) and two readouts (all coordinates or one random direction). The largest deviation from the law shrinks in proportion to p, from 0.08 at $p = 3\%$ to 0.003 at $p = 0.1\%$. The five-spin reservoir follows the law as well (Fig. 3c): at the largest memories computed, $s\,t_{\mathrm{mem}} \approx 20$, products of more than K inputs carry the share $e^{-K/(s\,t_{\mathrm{mem}})}$ of the capacity to within 0.07 with opposite phases and 0.14 with local loss.

B One qubit in closed form

This appendix solves the qubit of Sec. 3.3 exactly. It shows how mixing and the size of the stored signal depend on the turn per input and on the relaxation rate, and why the turn must shrink as the square root of the rate.

The model. The qubit relaxes towards $+X$ at a rate ϵ, through the jump $\sqrt{\epsilon}\,|+x\rangle\langle -x|$, and the input $u = \pm 1$ turns it about Z by $u\theta$ per unit time. Each input lasts one unit of time, and the qubit is read along Y. In the Bloch plane,

$$\dot{x} = -\epsilon(x - 1) - u\theta y, \qquad \dot{y} = u\theta x - \tfrac{1}{2}\epsilon y. \quad \text{(B1)}$$

Over one step this is a linear map of (x, y) plus an offset. For $u = +1$ the map has the form $\left(\begin{smallmatrix} a & -b \\ b & d \end{smallmatrix}\right)$ and the offset is (f, g); for $u = -1$ the signs of b and g flip. The input-averaged step is therefore diagonal, the stationary mean lies on the X axis

at $m = f/(1-a)$, and each input writes $u\,v$, with $v = bm + g$, along the measured axis Y.

How the orders arise. The averaged step keeps the two axes apart, while the input-dependent part swaps them. A signal written into Y must pass through X and back to pick up more inputs, so Y carries only odd orders. By the recursion (5), covariance passes from Y to X with the factor $b^2/(1-a^2)$ and back with $b^2/(1-d^2)$. Call the product of the two factors χ. Order $2j+1$ then carries the share $(1-\chi)\chi^j$ of the capacity of Y, and

$$\mathbb{E}D = \frac{1+\chi}{1-\chi}, \qquad \operatorname{Var} Y = \frac{v^2}{(1-d^2)(1-\chi)}; \tag{B2}$$

single past inputs carry the share $d^2(1-\chi)$.

Fixed turn. Let the relaxation slow down, $\epsilon \to 0$, at a fixed angle θ with $\sin\theta \neq 0$. Then $1-\chi = 3\epsilon/\sin^2\theta + O(\epsilon^2)$: the order grows as $1/\epsilon$, ϵD becomes exponentially distributed with rate $3/(2\sin^2\theta)$, and the share of single inputs vanishes, even for exact expectation values.

Gentle writing. If instead θ^2/ϵ approaches a constant ν, then to first order $a = 1-(1+\nu/2)\epsilon$ and $d = 1-(1+\nu)\epsilon/2$, with $b \approx \theta$ and $f \approx \epsilon$, and as $\epsilon \to 0$

$$\begin{aligned} \chi &\to \frac{\nu^2}{(\nu+2)(\nu+1)}, \\ \operatorname{Var} Y &\to \frac{4\nu}{(\nu+2)(3\nu+2)}. \end{aligned} \tag{B3}$$

The mean order stays finite, and single past inputs keep the share $1-\chi$. Their recall beyond delay τ is the fraction $d^{2\tau}$ of the total, so the range that holds 90% of it grows as $1/\epsilon$: $\epsilon\tau_{90} \to \ln 10/(1+\nu)$.

The best turn. The signal of Eq. (B3) is largest at $\nu = 2/\sqrt{3}$ (Fig. 3d). For $\nu \gg 1$ it vanishes as $4/(3\nu)$, because later inputs scramble earlier ones; for $\nu \ll 1$ it vanishes as ν, because each input writes too little. Among power laws $\theta \propto \epsilon^\alpha$, only $\alpha = 1/2$ keeps a finite signal: the turn must fall as the square root of the relaxation rate, that is, as one over the square root of the memory time. Direct propagation of the density matrix reproduces all these expressions.

C Three spins at strong loss

What we prove. Take three spins with one shared channel of opposite phases, Eq. (C1), the four Hamiltonians H1–H4 of Table 3, and every Hamiltonian whose three couplings lie in a small box around theirs. As γ grows:

- at a fixed input amplitude, the order of the memory of X_1, divided by γ, becomes exponentially distributed, and the signal in X_1 falls as $1/\gamma$ (boxes of half-width 1/32);
- with the amplitude scaled as $8/\sqrt{\gamma}$, single past inputs keep a finite share of the capacity of X_1, and the range of delays that holds 90% of their recall grows in proportion to γ (boxes of half-width 1/512).

Model and computations. The three spins evolve under

$$H_\eta(u) = \sum_{i<j} J_{ij} X_i X_j + \tfrac{1}{2}\sum_i Z_i + \tfrac{3+\eta u}{4}\sum_i X_i \tag{C1}$$

with one jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$, steps of $\Delta t = 1/2$ and signs $u = \pm 1$, at $\eta = 1$ (fixed amplitude) or $\eta = 8/\sqrt{\gamma}$ (gentle writing). The couplings of the four Hamiltonians H1–H4 are drawn from fixed seeds. For each loss rate we exponentiate the full generator for both signs, solve the stationary moment equations and evaluate Eq. (5) for X_1 until 99.99% of its memory is accounted for. The range τ_{90} is the smallest delay beyond which less than a tenth of the recall of single inputs remains.

Both regimes. At a fixed input amplitude the order of the memory grows in proportion to γ and single inputs lose their share (Fig. 6a), while the signal in a measured spin shrinks as $1/\gamma$. With the amplitude $\eta_0/\sqrt{\gamma}$ the order stays small (Fig. 6b), single inputs keep a finite share, the signal stays finite, and the range of delays over which single inputs are recalled grows in proportion to γ. How readable the memory is depends on the prefactor η_0 as on the turn for the qubit: at $\eta_0 = 8$ single inputs keep between 8% and 81% of the capacity of X_1 for the four Hamiltonians (Table 3), and for H3 this share rises to 67% at $\eta_0 = 2$, with a signal 25 times smaller.

Idea. At strong loss, one step of the full master equation is a rotation of the three dark states,

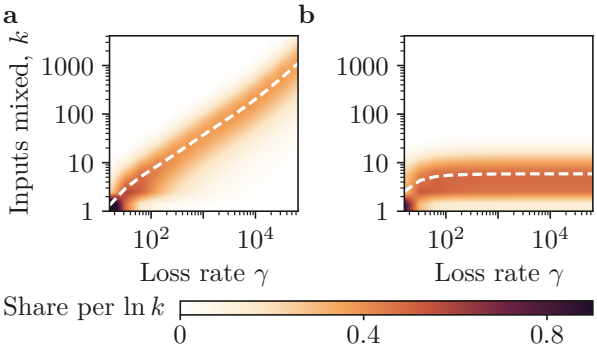


Figure 6: **Three spins, both regimes.** One shared channel of opposite phases, Hamiltonian H3: share of the capacity of X_1 carried by products of k inputs, per unit $\ln k$, at a fixed input amplitude (a) and with gentle writing, amplitude $8/\sqrt{\gamma}$ (b); dashed: mean order. Exact moments.

Table 3: Strong-loss limits for X_1 in the four three-spin reservoirs. Fixed amplitude: rate of the exponential law of the order divided by γ (certified). Gentle writing: share of single past inputs and range of their recall divided by γ, evaluated at $\gamma = 2^{16}$.

	rate	single-input share	range/γ
H1	78.6	0.81	0.50
H2	91.7	0.29	0.18
H3	61.4	0.08	0.24
H4	132.5	0.14	0.50

up to corrections of order $\epsilon = 1/\gamma$. At a fixed amplitude the two signs of the input rotate the dark states differently. The stored covariance then has a single slow direction, and the mixing law without a reset applies (Appendix A.5). With gentle writing both signs give almost the same rotation. The signal is written on the scale of the leak and collects in six slow directions, which we follow explicitly. Both arguments end in finitely many strict inequalities, which a computer checks with rigorous error bounds.

One step at strong loss. The jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$ leaves three dark states: the state with no excitation and the two single-excitation states with amplitudes $(0,1,-1)/\sqrt{2}$ and $(2,1,1)/\sqrt{6}$, which are orthogonal to the phases of the channel. It empties every other state, so at unit loss rate only the nine operators on the dark states do not decay. Let R embed these nine coordinates into the full operator space and $R^\flat$ project back along the decaying operators. Expanding one step in ϵ gives

$$E_u = R\, e^{\Delta t\, R^\flat \mathcal{H}_u R} R^\flat + \epsilon Z_u + O(\epsilon^2), \qquad \text{(C2)}$$

where $\mathcal{H}_u$ is the commutator with the Hamiltonian of input u. The first term rotates the dark states. The correction Z_u collects the brief excursions into bright states: one at the start of the step, one at its end, and the leak of Eq. (4) in between.

Fixed amplitude. At $\eta = 1$ the dark rotations O_u of the two signs differ, and the computer checks the conditions of the mixing law without a reset on the eight traceless dark coordinates. The average rotation forgets; the identity is the only covariance that the random rotations leave unchanged; and three constants from the first-order terms of Eq. (C2) are positive: the strength with which an input writes, the rate κ at which the slow covariance leaks, and the mixing fraction $s_0 = 1 - \|\mathbb{E}O_u\|_F^2/8$. The law then gives an exponential distribution of the order divided by γ, with rate κ/s_0 (Table 3), and the variance of X_1 falls as ϵ. A change r of the couplings moves each dark rotation by at most $3r$, which lets the checks cover the boxes of half-width $1/32$.

Gentle writing. At $\eta = 8\sqrt{\epsilon}$ both signs give the same leading rotation E, generated by the static part of the dark Hamiltonian. The input enters at order $\sqrt{\epsilon}$ and the leak at order ϵ. Expanding the step to this order gives the mean step $E + \epsilon E_1$ and the input-dependent part $u\sqrt{\epsilon}\,\Delta_1$.

Where the signal goes. The rotation E turns each of the three planes of coherence between the dark energy levels by its own angle, set by an energy gap, and the six phases of these turns differ from each other and from one. So E leaves only the populations unchanged, and on them E_1 generates a slow population dynamics with a unique stationary state m_*. An input therefore writes $\pm\sqrt{\epsilon}\,v$ with $v = \Delta_1 m_*$, which lies in the coherence planes: a new input first turns a population into a coherence.

Six slow directions. The covariances that E leaves unchanged form a six-dimensional space: the three covariances of the two population coordinates and one isotropic covariance in each coherence plane. Over $1/\epsilon$ steps a covariance in this space changes under two generators, both certified stable. K_B describes how the mean step

lets a stored signal decay, and K_T adds the random signs of later inputs, which turn a stored signal into products. With P_6 the projection onto the six directions, the single-input part of the stored covariance and the full covariance approach $W_* = -K_B^{-1} P_6(vv^\mathsf{T})$ and $\Sigma_* = -K_T^{-1} P_6(vv^\mathsf{T})$.

Limits. With a the coordinates of X_1, the share of single past inputs in the memory of X_1 tends to $a^\mathsf{T} W_* a / a^\mathsf{T} \Sigma_* a$. After t/ϵ steps the remaining single-input recall is the fraction $a^\mathsf{T} e^{tK_B} W_* a / a^\mathsf{T} W_* a$, which crosses 0.1 once; the crossing is the range divided by γ. The certificates show $a^\mathsf{T} W_* a > 0$, and since $\Sigma_* \succeq W_*$ the share is positive. In the boxes of half-width $1/512$ the six angles stay separated and both generators stay stable, so the limits stay positive.

Why coherence carries the memory. Returning bright population to the dark subspace after every step leaves the leading terms of the scaled step unchanged, so it keeps the same limits. Dephasing in the dark energy basis, by contrast, removes the coherence planes into which v writes. Restricted to populations, the two signs then differ only at order $\epsilon^{3/2}$, because the response of a population at order $\sqrt{\epsilon}$ is a coherence, while the populations still follow the slow population dynamics. A new input thus writes a vector of size $\epsilon^{3/2}$, the slow dynamics sum its square over $1/\epsilon$ steps, and the stored signal is of order ϵ^2: the collapse as $1/\gamma^2$ of Sec. 3.3. Numerically we apply the loss-only projector $\mathcal{P}$ after every step; one version keeps the recovered state, the other also dephases it in the energy basis of the projected Hamiltonian. From $\gamma = 2^8$ to 2^{18} the variance of X_1 stays between 10^{-4} and 2×10^{-3} when the coherences are kept and falls as γ^{-2}, by six orders of magnitude, when they are erased; at a fixed input amplitude it falls as $1/\gamma$.

What the computer checks. The attached archive `spin_certificates.zip` contains the generators, the exact couplings and the interval programs. They enclose every quantity above in intervals with rigorous rounding (ball arithmetic of Arb, python-flint 0.9.0) and check each strict inequality: at a fixed amplitude, that the average rotation forgets, that the identity is the only fixed covariance and that the writing strength, κ and s_0 are positive; for gentle writing, that the six angles are separated, that both generators are stable, that $a^\mathsf{T} W_* a$ is positive (in the archive, the numerator of an auxiliary recall quantity) and where the remaining recall crosses 0.1. At 192 bits they reproduce the archived enclosures, and at 256 bits they check every strict inequality. A second program, sharing no routine with them (see Author contributions), builds the full finite-rate maps and extrapolates to infinite loss; it agrees with the certified rates and limiting recalls to better than one part in a million.

D Numerical details and supporting checks

This appendix collects what is needed to reproduce the five-spin results and the checks behind several statements of the main text, in the order of the Results.

Recall and memory (Sec. 3.1). Where the figures draw C_τ against the delay, each delay is averaged with its neighbours, with weights that fall linearly to zero at a fifth of the delay; this removes the noise of the finite test set and enters no sum. The maps of Fig. 2a,b are computed at $\gamma = 0.1$, 0.25 and 2^{-1} to 2^6 in powers of two and interpolate between them by shifting each recall profile along the delay. At a fixed amplitude the recall of one and two shared channels reaches past the 300 delays summed beyond $\gamma = 64$, so Fig. 2c shows them up to $\gamma = 64$; summed over all delays with exact moments (Sec. 5.4), one shared channel keeps 27 to 35 units and two channels 26 to 30 from $\gamma = 64$ to 256 on two reservoirs. The recall of opposite phases and pairwise loss ends well within the window up to $\gamma = 256$. With gentle writing, delays beyond 300 would add about 1% to the memory of opposite phases.

Dark states (Sec. 3.1). For the dark-state model we write $\mathcal{P} = RR^\flat$ with $R^\flat R = 1$, where R embeds the 100 dark coordinates α, evolve α under Eq. (4) and record the same 45 features of the full state $(R - \mathcal{S}\mathcal{H}(u)R/\gamma)\,\alpha$. The readout is trained on these features exactly as for the full reservoir; the model itself has no fitted parameter (Fig. 7a). For the erasure test we replace the dark coordinates after every step by their time average, $r \mapsto r - RR^\flat(r - \bar{r})$; the control erases the dark subspace rotated by a random product of single-spin unitaries. Recall is then measured at delays

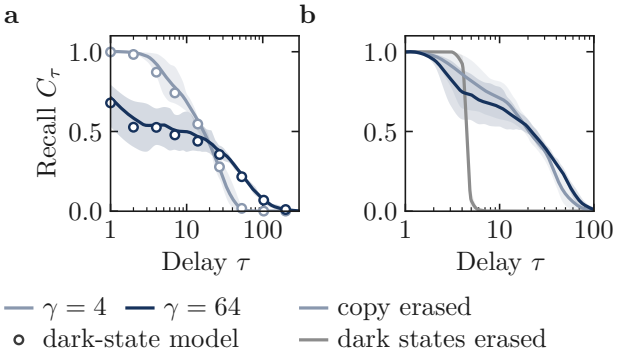


Figure 7: **Dark states hold the memory.** One shared channel. **a** Recall of the full reservoir and of the dark-state model, with no fitted parameter. **b** Recall at $\gamma = 16$ of the intact reservoir (dark line) and when the dark coordinates, or a copy of the dark subspace rotated by random single-spin unitaries, are erased after every step.

1 to 100 with 800 training, 200 validation and 800 test inputs (Fig. 7b).

For one shared channel $\gamma\bar{\kappa}$ approaches $\kappa_c = 2.945$, to four digits from $\gamma = 512$ to 16384, so t_f grows in proportion to γ, while from $\gamma = 4096$ to 16384 it changes only from 56 to 58 steps with opposite phases and from 103 to 114 with pairwise loss. The second-order calculation of Eq. (4) gives the same κ_c at five spins and 1.4, 2.1 and 3.4 at three, four and six spins: the leak coefficient grows only slowly with the size.

Time to forget and phases (Sec. 3.2). We compute t_f at $\eta = 1$, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$, for γ from 2^{-8} to 2^6 with local loss and from 2^{-2} to 2^{12} with opposite phases, on eight reservoirs. With the slowest mode in place of the mean rate, the curves of Fig. 3a collapse as well, to within 11% wherever forgetting takes more than four steps (3% beyond $\gamma\eta^2 = 32$). With the input on spin 1 only (four reservoirs) the collapses hold to within a few percent. On one reservoir with exact moments, the memory peak of local loss moves from $\gamma \approx 1/16$ at $\eta = 1$ to 1/64 at $\eta = \frac{1}{8}$, while the best memory rises from 16 to 39 units; the mean over eight reservoirs at $\eta = 1$ is 14.5 (Table 2).

The write per input into the dark states (Fig. 3b) is $2\|R^\flat[E(\frac{3}{4}) - E(\frac{1}{4})]r_*\|$ at the stationary state r_* of the equal-phase channel, for the input patterns $(1, \ldots, 1)$ and $\sqrt{2}\cos(2\pi i/5)$ multiplying X_i. Along the phases it falls as $1/\gamma$, to 0.003 at $\gamma = 128$; across them it falls only slowly, to 0.18. Opposite phases are the equal-phase channel with the field on spins 4 and 5 reversed, up to a relabelling of the random couplings (a turn of these spins by π about Z).

The range τ_{90} of the exact linear recall grows with the clock but not in proportion: along the gentle line of opposite phases, where $\gamma\eta^2$ is fixed and t_f grows as γ, it grows from 32 steps at $\gamma = 16$ to 155 at $\gamma = 256$, about as $\gamma^{0.6}$, while the memory approaches the 45 features.

Orders and capacities (Secs. 3.2 and 3.3). For the five-spin reservoir the expectations over the input in Eq. (5) use six-point Gauss–Legendre quadrature, and the equations $X - \bar{A}X\bar{A}^\mathsf{T} = C$ are solved in the eigenbasis of $\bar{A}$. The shares of the 45 features are summed until the rest falls below 10^{-8}, or until successive orders shrink at a settled ratio, whose geometric tail we add. The linear recall follows from the same moments, $C_\tau = c_\tau^\mathsf{T}\Sigma_y^{-1}c_\tau/\mathrm{Var}\,u$ with c_τ the covariance of the features with $u_{t-\tau}$ and Σ_y their covariance; it reproduces the simulated memories to a median of 2%, and where much of the recall is faint, as for opposite phases at $\gamma = 256$ and the full amplitude, it finds up to a fifth more than the finite training set of the simulations. Three steps of inverse iteration on $\mathcal{T}$, from the input's covariance V and from the identity, give t_{mem}, Q and ℓ. Loss rates run in powers of four, at the full amplitude from 1 to 4096 for opposite phases and from 4 down to 1/1024 for local loss, and with gentle writing to 16384 and 1/4096, on four reservoirs (two at the two largest memory times); the control at $\gamma = 1$ runs down to $\eta = 1/32$ on two reservoirs, and local loss at $\eta = \frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ on one. With local loss at the full amplitude the mean order stays within 2.2 of $s\,t_{\mathrm{mem}}$ up to $s\,t_{\mathrm{mem}} = 18$ ($\gamma = 1/1024$).

Capacities by polynomial degree follow Ref. [34]: products of Legendre polynomials in the past inputs $2u - 1$, up to total degree four within finite windows of delays, fitted by the same features and kept above a noise floor from shuffled inputs. At $\gamma = 1$ degrees three and four carry 8.2 units with one shared channel, 15.9 with opposite phases and 16.4 with local loss.

We measure the size of the signal by the variance of a feature over the test inputs, averaged over the features: with opposite phases about 5×10^{-6} with gentle writing from $\gamma = 16$ to 256 and 8×10^{-5} at the full amplitude; with local loss and gentle writing 7×10^{-5} at $\gamma = 1/32$; for the equal-phase channel at the full amplitude from

2×10^{-4} at $\gamma = 1$ to 4×10^{-7} at $\gamma = 64$.

Sampling and polarization (Sec. 3.4). Eight times more training inputs change the memory at 10^6 and 10^8 preparations by less than 1.3 units. At 10^{10} opposite phases with gentle writing peak near $\gamma = 64$ with 22 units, and at 10^8 weak local loss with gentle writing keeps 19. At 10^4 preparations the full amplitude keeps 9 units with local loss and 5 with opposite phases, against 4 and 1 with gentle writing; with the input on a single spin, which writes more gently by itself, the crossings move to about 3×10^6 and 6×10^7 preparations (four reservoirs).

For the matched clock we find, for every shape and reservoir, the loss rate at which $t_{\mathrm{leak}} = 32$ steps, at weak loss without protected operators and at strong loss with them, and run the reservoir there with $\eta = \sqrt{4/32}$. The write is $-i[H_{\mathrm{in}}, \rho_*]$ with $H_{\mathrm{in}} = h \sum_i X_i$ and ρ_* the stationary state, projected spectrally onto the slow modes: the protected ones, or without protected operators those decaying at most four times faster than the slowest. The memory that survives 10^6 preparations rises with the write, from 0.3 units for the shared channels to 17 for local loss.

Ordinary noise (Sec. 4). Adding local loss $\sqrt{\gamma_{\mathrm{loc}}}\,\sigma_i^-$ to a shared channel gives $\bar{\kappa} = \kappa_{\mathrm{c}}/\gamma + \kappa_{\mathrm{l}}\gamma_{\mathrm{loc}}$, where to first order κ_{l} is the trace of the added dissipator on the 99 slow dark modes divided by 99, $140/99 \approx 1.414$; local dephasing $\sqrt{\gamma_\phi}\,\sigma_i^z$ adds $\kappa_\phi \gamma_\phi$ with $\kappa_\phi = 403.2/99 \approx 4.073$. For γ from 32 to 256 and noise rates from 0.001 to 0.1 the full spectra reproduce both coefficients to within 0.2%. A time to forget of at least T steps, $\kappa_{\mathrm{c}}/\gamma + \kappa_{\mathrm{l}}\gamma_{\mathrm{loc}} \le 1/(\Delta t\, T)$, then needs $\gamma/\gamma_{\mathrm{loc}} \ge 4\kappa_{\mathrm{c}}\kappa_{\mathrm{l}}\,\Delta t^2 T^2 \approx 4.2\,T^2$, or $12\,T^2$ for dephasing.

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